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结合单目深度估计与立体匹配释放潜能

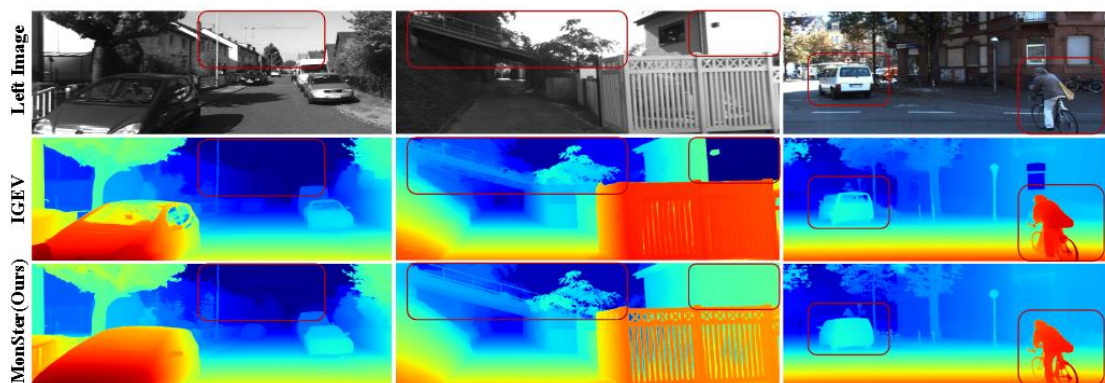


图 1 零样本泛化比较。所有的模型都在 SceneFlow 上训练，在 KITTI 上测试。相比于基线模型 IGEV^[38]，我们的方法 MonSter 展现出了明显的在挑战性区域的效果改善(比如反射表面，无纹理区域，细小结构，以及遥远的物体)

摘要

立体匹配通过图像对应关系来恢复深度。现有的方法在缺乏匹配线索的病态区域比如遮挡和无纹理区域往往表现不佳。为了解决这个问题，我们提出了 MonSter，这是一种利用单目深度估计和立体匹配互补优势的新颖方法。MonSter 将单目深度和立体匹配集成到一个双分支架构中，以迭代地相互改进。基于置信度的引导机制自适应地选择可靠的立体线索来进行单目深度的尺度-偏移恢复。反过来，细化后的单目深度又能在病态区域有效地指导立体匹配。这种迭代式的相互增强使得 MonSter 能够将单目深度先验从粗糙的对象级结构演变为像素级的几何结构，从而充分释放立体匹配的潜力。如图 2 所示，MonSter 在五个最常用的排行榜（SceneFlow、KITTI 2012、KITTI 2015、Middlebury 和 ETH3D）上均排名第一。在 ETH3D 上，相比之前的最佳方法，提升幅度高达 49.5%。综合分析验证了 MonSter 在病态区域的有效性。在零样本泛化方面，MonSter 在各项指标上均显著且持续地优于最先进的方法。代码已公开发布：<https://github.com/Junda24/Monster>。

1 简介

立体匹配通过校正后的立体图像对估计视差图，随后可将其转换为度量深度。它是自动驾驶、

机器人导航和 3D 重建等许多应用的核心。

基于深度学习的方法^{[4][14][16][20][35][48]}在标准基准测试中已展现出令人鼓舞的性能。这些方法大致可分为基于代价过滤的方法和基于迭代优化的方法。基于代价过滤的方法^{[4][14][15][22][37][39][41][43]}使用 CNN 特征构建 3D/4D 代价体，随后进行一系列 2D/3D 卷积以进行正则化和过滤，从而最小化错误匹配。基于迭代优化的方法^{[10][16][20][38][49]}最初构建全对相关体，然后索引局部代价以提取运动特征，这些特征指导循环单元(ConvGRUs)^[9]迭代地细化视差图。

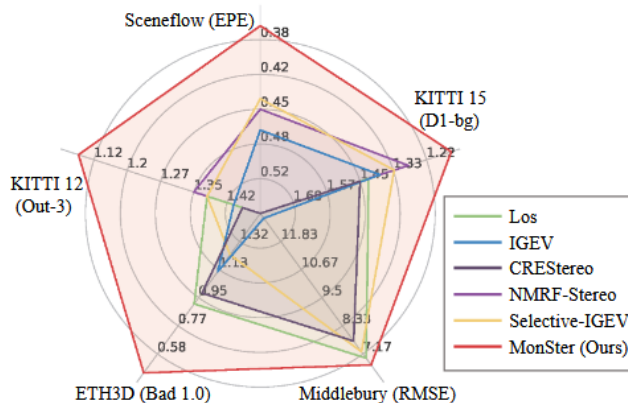


图 2 排行榜单表现。我们的方法在 5 个排行榜中都排行第一，大幅提升了 SOTA

这两类方法本质上都是通过相似性匹配推导视差，假设两幅图像中都存在可见的对应关系。这在匹配线索有限的病态区域（例如遮挡、无纹理区域、重复/细微结构以及像素表现较低的远距离物体）构成了挑战。现有方法^{[4][31][33][37][42][44][49]}通过在特征提取或代价聚合过程中增强特征表示来解决这一问题：GMStereo^[42]和 STTR^[18]采用 Transformer 作为特征提取器；IGEV^[38]和 ACVNet^[37]利用注意力机制将几何信息整合到代价体中以增强匹配信息；Selective-IGEV^[35]和 DLNR^[49]通过在迭代细化阶段引入高频信息进一步提高了性能。然而，这些方法仍然难以从根本上解决误匹配问题，限制了其实际性能。

与立体匹配不同，单目深度估计直接从单张图像恢复 3D 信息，不会遇到误匹配的挑战。虽然单目深度为立体匹配提供了互补的结构信息，但预训练模型通常产生的是带有尺度和偏移模糊的相对深度。如图 3 所示，单目深度模型的预测与真值差异很大。即使经过全局尺度和偏移对齐，仍然存在大量误差，使得单目深度与立体视差的像素级融合变得复杂。基于这些见解，我们提出了 MonSter，这是一种新颖的方法，它将立体匹配分解为单目深度估计和像素级尺度-偏移恢复，充分结合了单目和立体算法的优势，并克服了缺乏匹配线索的限制。

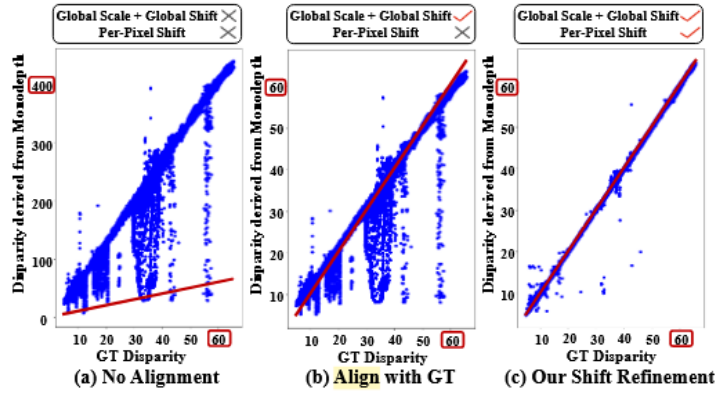


图 3 真实视差与 KITTI 数据集中单目逆深度^[46]之间的差距。红线代表相同的视差图。
 (a): 无对齐。(b): 使用全局尺度和全局位移值(对所有像素都是一样的)对齐深度与 GT。
 (c): Monster 产生的单目视差, 逐像素位移细化。即使与 GT 全局对齐, SOTA 单目深度模型仍然表现出严重的噪声。我们的方法 Monster 有效地解决了这一问题, 通过逐像素偏移细化单目深度, 充分释放了单目深度先验在立体匹配任务中的潜力

MonSter 为单目深度估计和立体匹配构建了单独的分支, 并通过立体引导对齐和单目引导细化模块自适应地融合它们。SGA 首先通过与立体视差进行全局对齐, 将单目深度重新缩放为单目视差。然后, 它使用条件引导的 GRU 自适应地选择可靠的立体线索来更新像素级的单目视差偏移。作为 SGA 的对称操作, MGR 使用优化后的单目视差作为条件, 在匹配失败的区域自适应地细化立体视差。通过多次迭代, 两个分支有效地相互补充: (1)虽然单目深度在粗糙的对象级别上有益, 但直接单向地将单目深度融合到立体匹配中会受到尺度-偏移模糊的影响, 这通常会在倾斜或弯曲表面等复杂区域引入噪声。利用立体信息细化单目深度有效地解决了这个问题, 确保了 MonSter 的鲁棒性。(2)细化后的单目深度反过来在具有挑战性的区域为立体匹配提供了强有力的指导。例如, 立体匹配的深度感知能力会随着距离的增加而下降, 这是由于像素占比变小和匹配难度增加所致。在大规模数据集上预训练的单目深度模型受此类问题的影响较小, 可以有效地改善相应区域的立体视差。

我们的主要贡献总结如下:

- 我们提出了一种新颖的立体匹配方法 MonSter, 充分利用像素级单目深度先验, 显著提高了立体匹配在病态区域和精细结构中的深度感知性能。
- 如图 2 所示, MonSter 在 5 个广泛使用的排行榜上排名第一: KITTI 2012^[11]、KITTI 2015^[23]、Scene Flow^[22]、Middlebury^[27]和 ETH3D^[28]。将现有最先进技术 (SOTA) 提升了高达 49.5%。

- 与 SOTA 方法相比, MonSter 在各种数据集中始终实现最佳的零样本泛化。仅在合成数据上训练的 MonSter 在各种真实世界数据集中表现出强大的性能(见图 1)。

2 相关工作

基于匹配的方法。主流立体匹配方法通过匹配代价恢复视差。这些方法通常分为两类：基于代价过滤的方法和基于迭代优化的方法。基于代价过滤的方法^{[5][6][12][14][22][29][43]}通常使用特征图构建 3D/4D 代价体,随后采用 2D/3D CNN 过滤该体积并推导最终视差图。构建具有强表示能力的代价体以及从噪声代价体中准确回归视差是关键挑战。PSMNet^[4]提出了堆叠沙漏 3D CNN 以实现更好的代价正则化。^{[14][30][37]}其他方法分别提出了分组相关体、注意力连接体和金字塔变形体,以提高匹配代价的表达能力。最近,基于迭代优化的一类新方法^{[16][20][35][38][49]}在准确性和效率方面都取得了最先进的性能。这些方法使用 ConvGRU 利用从全对相关体中检索到的局部代价值来迭代更新视差。同样,基于迭代优化的方法也主要集中在改进匹配代价构建和迭代优化阶段。CREStereo^[16]提出了一种级联循环网络,以从粗到细的方式更新视差场。IGEV^[38]提出了几何编码体,编码几何和上下文信息以获得更鲁棒的匹配代价。这两种类型的方法本质上都是从匹配代价中推导深度,因此在病态区域受到固有这种机制的限制。

结合结构先验的立体匹配。在病态区域进行匹配具有挑战性,旧方法^{[7][17][19][31][44][47]}试图利用结构先验来解决这个问题。EdgeStereo^[31]通过将边缘检测线索整合到视差估计流程中来增强边缘区域的性能。SegStereo^[44]利用分割网络的语义线索作为立体匹配的引导,提高了无纹理区域的性能。然而,语义和边缘线索仅提供对象级先验,不足以进行像素级深度感知。因此,它们在具有大曲面或倾斜表面的复杂场景中效果不佳。因此,一些方法结合单目深度先验,通过密集相对深度为立体匹配提供逐像素指导。CLStereo^[47]引入了一个单目分支作为上下文约束,将几何先验从单目分支转移到立体分支。Los^[17]使用单目深度作为局部结构先验来生成倾斜平面,这可以显式利用结构信息来更新视差。然而,如图 3 所示,单目深度估计存在严重的尺度和偏移模糊。直接将其作为结构先验来约束立体匹配可能会引入严重的噪声。我们的 MonSter 自适应地选择可靠的立体视差来校正单目深度的尺度和偏移,这既充分利用了单目深度先验,又避免了噪声,从而显著增强了立体匹配在病态区域的性能。

3 方法

如图 4 所示，MonSter 由 1)单目深度分支，2)立体匹配分支，和 3)相互细化模块组成。这两个分支估计初始单目深度和立体视差，它们被输入到相互细化模块中以迭代地相互改进。

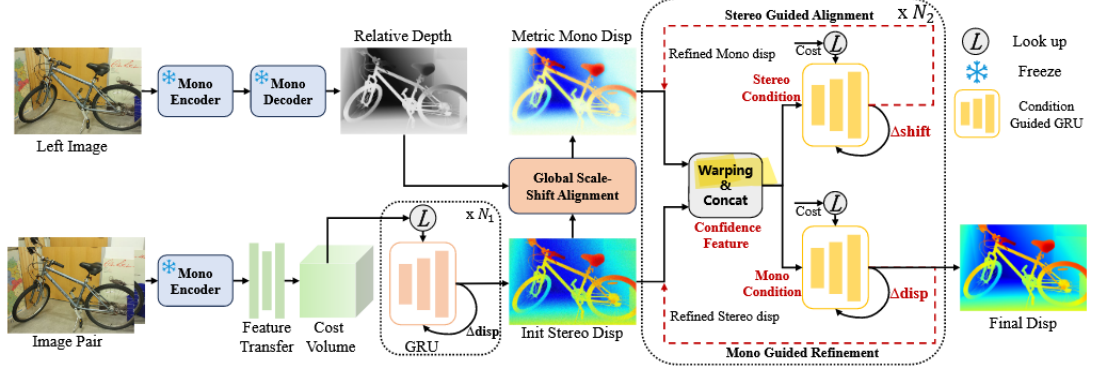


图 4 **Monster 概览**。Monster 由单目深度估计分支、立体匹配分支和相互微调模块组成。

它利用另一个分支的先验信息，迭代地改进一个分支，有效地解决了立体匹配中的不适定性问题

3.1 单目和立体分支

单目深度分支可以利用大多数单目深度模型来实现显著的性能提升（分析见第 4.5 节）。我们最佳的经验配置使用预训练的 DepthAnything V2^[46]作为单目深度分支，该分支使用 DINOv2^[24]作为编码器，DPT^[26]作为解码器。立体匹配分支遵循 IGEV^[38]以获得初始立体视差，仅对特征提取组件进行了修改，如图 4 所示。为了高效且充分地利用预训练的单目模型，立体分支与单目分支共享 DINOv2 中的 ViT 编码器，参数被冻结以防止立体匹配训练影响其泛化能力。此外，ViT 架构以单一分辨率提取特征，而最近的立体匹配方法通常利用四种尺度的多尺度特征（原始图像分辨率的 $\frac{1}{32}, \frac{1}{16}, \frac{1}{8}, \frac{1}{4}$ ）。为了与 IGEV 完全对齐，我们采用一堆 2D 卷积层，称为特征传输网络，对 ViT 特征进行下采样和转换，生成金字塔特征集合 $\mathcal{F} = \{F_0, F_1, F_3, F_4\}$ 。我们遵循 IGEV 构建几何编码体，并使用相同的 ConvGRU 进行迭代优化。为了平衡准确性和效率，我们仅执行 N_1 次迭代以获得质量合理的初始立体视差。

3.2 相互细化

一旦获得初始（相对）单目深度和立体视差，它们就被输入到相互细化模块中进行迭代相互改进。我们首先执行全局尺度-偏移对齐，将单目深度转换为视差图，并使其与立体输出粗略对齐。然后我们迭代执行双分支细化：立体引导对齐（SGA）利用立体线索更新单目视差的逐像素偏移；单目引导细化（MGR）利用对齐的单目先验进一步细化立体视差。

全局尺度-偏移对齐 (Global Scale-Shift Alignment)。 全局尺度-偏移对齐对全局尺度和全局偏移执行最小二乘优化，以将逆单目深度与立体视差粗略对齐：

$$s_G, t_G = \underset{s_G, t_G}{\operatorname{argmin}} \sum_{i \in \Omega} (s_G D_M(i) + t_G - D_S^0(i))^2 \quad (1)$$

$$D_M^0 = s_G D_M + t_G$$

其中 $D_M(i)$ 分别 $D_S^0(i)$ 是第 i 个像素处的逆单目深度和立体视差。 Ω 代表立体视差值按升序排列在 20% 到 90% 之间的区域，这有助于过滤不可靠的区域，如天空、极远区域和近距离异常值。直观地说，此步骤将逆单目深度转换为与立体视差粗略对齐的视差图，从而能够在同一空间中进行有效的相互细化。在本文的其余部分，我们将此对齐的视差图 D_M^0 称为单目视差。

立体引导对齐 (SGA)。 虽然经过了粗略对齐，但统一的偏移 t_G 不足以恢复不同像素处的准确单目视差。为了充分释放单目深度先验的潜力，SGA 利用中间立体线索进一步估计逐像素残差偏移。为了避免引入嘈杂的立体线索，SGA 使用基于置信度的引导。在每个更新步骤 j 中，我们使用流残差图 (flow residual map) F_S^j 计算置信度，该图是通过基于立体视差 D_S^j 变形 (warping) 和减去特征获得的：

$$F_S^j(x, y) = \|F_3^L(x, y) - F_3^R(x - D_S^j, y)\| \quad (2)$$

其中 F_3^L 和 F_3^R 分别代表左右图像的四分之一分辨率特征。对于每次迭代，我们还使用当前立体视差 D_S^j 从几何编码体中索引以获得立体分支的几何特征 G_S^j （遵循 IGEV）。与 F_S^j 和 D_S^j 拼接后，我们获得立体条件线索 x_S^j ：

$$x_S^j = [En_g([G_S^j, F_S^j, D_S^j]), En_d(D_S^j), D_S^j] \quad (3)$$

其中 En_g 和 En_d 是两个用于特征编码的卷积层。我们将 x_S^j 输入到条件引导的 ConvGRU 中以更新单目分支的隐藏状态 h_M^{j-1} ：

$$\begin{aligned}
 z^j &= \sigma(\text{Conv}([h_M^{j-1}, x_S^j], W_z) + c_k), \\
 r^j &= \sigma(\text{Conv}([h_M^{j-1}, x_S^j], W_r) + c_r), \\
 \tilde{h}_M^j &= \tanh(\text{Conv}([r^j \odot h_M^{j-1}, x_S^j], W_h) + c_h), \\
 h_M^j &= (1 - z^j) \odot h_M^{j-1} + z^j \odot \tilde{h}_M^j
 \end{aligned} \tag{4}$$

其中 c_k, c_r, c_h 是上下文特征。基于隐藏状态 h_M^j ，我们通过两个卷积层解码残差偏移 Δ 以更新单目视差：

$$D_M^{j+1} = D_M^j + \Delta t \tag{5}$$

单目引导细化 (MGR)。 与 SGA 对称，MGR 利用对齐的单目视差来解决立体匹配在病态区域、细微结构和远距离物体中的不足。具体来说，我们采用具有独立参数的相同条件引导 GRU 架构来细化立体视差。我们同时计算单目和立体分支的流残差图 F_M^j, F_S^j 和几何特征 G_M^j, G_S^j ，提供全面的立体细化指导：

$$\begin{aligned}
 F_M^j(x, y) &= \|F_3^L(x, y) - F_3^R(x - D_M^j, y)\|_1 \\
 x_M^j &= [\text{En}_g([G_M^j, F_M^j, D_M^j]), \text{En}_d(D_M^j), D_M^j, \text{En}_g([G_S^j, F_S^j, D_S^j]), \text{En}_d(D_S^j), D_S^j]
 \end{aligned} \tag{6}$$

其中 G_M^j 是通过使用单目视差 D_M^j 从几何编码体索引获得的单目分支的几何特征。 x_M^j 代表 ConvGRU 的单目条件，我们类似地使用公式(4)更新立体隐藏状态 h_S^j ，仅将条件输入从 x_S^j 更改为 x_M^j 。我们使用相同的两个卷积层来解码残差视差 Δ 并遵循公式(5)更新当前立体视差。经过 N^2 轮双分支细化后，立体分支的视差即为 MonSter 的最终输出。

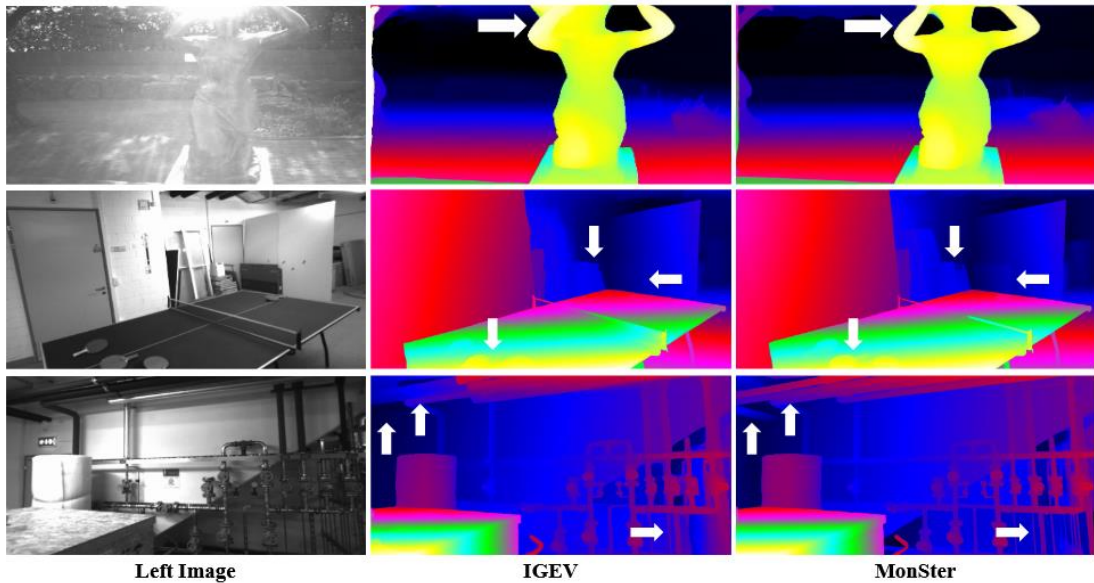


图 5 ETH3D 上的定性结果 Monster 在具有强反射、精细结构等挑战性区域地表现优于 IGEV

3.3 损失函数

我们使用 L1 损失来监督两个分支的输出。我们将立体分支前 N_1 次迭代的视差集表示为 $\{d_i\}_{i=0}^{N_1-1}$ ，并遵循[20]随着迭代次数的增加呈指数增加权重。总损失定义为单目分支损失 $\mathcal{L}_{Mono}$ 和立体分支损失 $\mathcal{L}_{Stereo}$ 之和，如下所示：

$$\begin{aligned}\mathcal{L}_{Stereo} &= \sum_{i=0}^{N_1-1} \gamma^{N_1+N_2-i} \|d_i - d_{gt}\|_1 + \sum_{i=N_1}^{N_1+N_2-1} \gamma^{N_1+N_2-i} \|d_i - d_{gt}\|_1 \\ \mathcal{L}_{Mono} &= \sum_{i=N_1}^{N_1+N_2-1} \gamma^{N_1+N_2-i} \|D_m^{i-N_1} - d_{gt}\|_1\end{aligned}\quad (7)$$

其中 $\gamma = 0.9$ ， d_{gt} 是真值。

4 实验

4.1 实现细节

我们使用 PyTorch 实现 MonSter，并使用 NVIDIA RTX 3090 GPU 进行实验。我们使用 AdamW^[21]优化器，并将梯度裁剪到 $[-1, 1]$ 范围内，遵循基线^[38]。我们使用单周期学习率计划，学习率为 $2e-4$ ，并将 MonSter 以批量大小 8 训练 200k 步作为预训练模型。对于单目分支，我们使用 DepthAnything V2^[46]的 ViT-large 版本，并冻结其参数以防止立体匹配任务的训练影响其泛化和准确性。

遵循标准做法^{[16][17][35]}，我们在 Scene Flow^[22]上预训练 MonSter 以进行大多数实验。为了在 ETH3D 和 Middlebury 上进行微调，我们遵循 SOTA 方法^{[16][17][35]}，从各种公共数据集中创建基础训练集（BTS）进行预训练，包括 Scene Flow^[22]、CREStereo^[16]、Tartan Air^[34]、Sintel Stereo^[3]、Falling Things^[32]和 InStereo2k^[2]。

表 1 在 SceneFlow 测试集上的定量评估。最好的结果加粗，次好的结果下划线标记

方法	GwcNet	LEAStereo	ACVNet	IGEV	Selective-IGEV	Monster
EPE (px)	0.76	0.78	0.48	0.47	<u>0.44</u>	0.37 _(-15.91%)

表 2 常用基准测试上的结果。所有成绩均来自官方排行榜出版物或相关论文。所有指标均以百分比表示，只有 RMSE 以像素为单位。对于测试掩码，“All”表示用所有像素测试，“Noc”表示用无遮挡区域测试。**最好的结果加粗**，次好的结果下划线标记。

Method	ETH3D ^[28]			Middlebury ^[27]			KITTI 2015 ^[11]				KITTI 2012 ^[11]			
	Bad10	Bad1.0	RMSE	Bad2.0	Bad2.0	Bad2.0	D1-bg	D1-all	D1-bg	D1-all	Out-2	Out-2	Out-3	Out-3
	Noc	All	Noc	Noc	All	Noc	Noc	Noc	All	All	Noc	All	Noc	All
GwcNet ^[14]	6.42	6.95	0.69	-	-	-	1.61	1.92	1.74	2.11	2.16	2.71	1.32	1.70
GANet ^[48]	6.22	6.86	0.75	-	-	-	1.40	1.73	1.55	1.93	1.89	2.50	1.19	1.60
LEAStereo ^[8]	-	-	-	7.15	12.10	8.11	1.29	1.51	1.40	1.65	1.90	2.39	1.13	1.45
ACVNet ^[37]	2.58	2.86	0.45	13.70	19.50	32.2	1.26	1.52	1.37	1.65	1.83	2.35	1.13	1.47
RAFT-Stereo ^[20]	2.44	2.60	0.36	4.74	9.37	8.41	1.44	1.69	1.58	1.82	1.92	2.42	1.30	1.66
CREStereo ^[16]	0.98	1.09	<u>0.28</u>	3.71	8.13	7.70	1.33	1.54	1.45	1.69	1.72	2.18	1.14	1.46
IGEV ^[38]	1.12	1.51	0.34	4.83	8.16	12.80	1.27	1.49	1.38	1.59	1.71	2.17	1.12	1.44
CroCo-Stereo ^[36]	0.99	1.14	0.30	7.29	11.11	8.91	1.30	1.51	1.38	1.59	-	-	-	-
DLNR ^[49]	-	-	-	3.20	6.98	7.78	1.42	1.61	1.60	1.76	-	-	-	-
Selective-IGEV ^[35]	1.23	1.56	0.29	2.51	6.04	7.26	1.22	<u>1.44</u>	1.33	<u>1.55</u>	<u>1.59</u>	<u>2.05</u>	1.07	1.38
LoS ^[17]	<u>0.91</u>	<u>1.03</u>	0.31	4.20	8.03	<u>6.99</u>	1.29	1.52	1.42	1.65	1.69	2.12	1.10	1.38
NMRF-Stereo ^[13]	-	-	-	-	-	-	<u>1.18</u>	1.46	<u>1.28</u>	1.57	<u>1.59</u>	2.07	<u>1.01</u>	<u>1.35</u>
MonSter(我们的)	0.46	0.72	0.20	<u>2.64</u>	<u>6.14</u>	6.71	1.05	1.33	1.13	1.41	1.36	1.75	0.84	1.09

4.2 基准性能

为了证明我们方法的出色性能，我们在五个最常用的基准上评估了 MonSter: KITTI 2012^[11]、KITTI 2015^[23]、ETH3D^[28]、Middlebury^[27]和 Scene Flow^[22]。

Scene Flow^[22]。我们在 Scene Flow 上取得了新的最先进性能，EPE 指标为 0.37，比我们的基线^[38]提升了 21.28%，并优于 SOTA 方法^[35]15.91%。

ETH3D^[28]。遵循 SOTA 方法^{[16][17][35][40][42]}，完整的训练集由 BTS 和 ETH3D 训练集组成。我们的 MonSter 在 ETH3D 排行榜上排名第一。MonSter 在所有报告的指标中均取得了所有已发布方法中的最佳性能。与基线 IGEV 相比，我们在三个报告的指标中分别实现了 58.93%、52.32% 和 41.18% 的改进。定性比较显示出类似的趋势。值得注意的是，即使与之前的最佳方法 LoS 相比，我们也从 0.91 提高到了 0.46 的 Bad 1.0 (Noc) 指标，实现了 49.45% 的改进。

Middlebury^[27]。同样^{[16][17][35][40][42]}，Middlebury 的训练集是 BTS 和 Middlebury 训练集的组合。MonSter 在 RMSE 指标方面优于所有现有方法。与基线^[38]相比，我们在 Bad 2.0 (Noc) 指标上实现了 45.34% 的改进。

KITTI。遵循 SOTA 方法^{[13][16][35]}的训练，我们在 KITTI 2012 和 KITTI 2015 的混合数据集上对 Scene Flow 预训练模型微调了 50k 步。在撰写本文时，MonSter 在 KITTI 2012 和 KITTI 2015 排行榜上同时排名第一。就像表 2 中展示的。我们在所有指标上都取得了最佳性能。在 KITTI 2015 上，MonSter 在所有区域的 D1-all 指标上分别超过 CREStereo^[16]和 Selective-IGEV^[35]16.57% 和 9.03%。至于 KITTI 2012，我们大幅超越了现有的 SOTA。与目前精度最高的方法 NMRF-Stereo^[13]和 Selective-IGEV^[35]相比，我们在所有区域的 Out-3 指标上分别实现了 19.26% 和 21.01% 的改进。

表 3 在 KITTI2012 排行榜反射区域的测试结果。最好的结果加粗。

	KITTI 2021 Refleticive Region					
	Out-2	Out-2	Out-3	Out-3	Out-4	Out-4
	Noc	All	Noc	All	Noc	All
ACVNet ^[37]	11.42	13.53	7.03	8.67	5.18	6.48
CREStereo ^[16]	9.71	11.26	6.27	7.27	4.93	5.55

IGEV ^[38]	7.57	8080	4.35	5.00	3.16	3.57
Selective-IGEV ^[35]	6.73	7.84	3.79	4.38	2.66	3.05
LoS ^[17]	6.31	7.84	3.47	4.45	2.41	3.01
NMRF-Stereo ^[13]	10.02	12.34	6.35	8.11	4.80	6.09
Monster(我们的)	5.66	6.81	2.75	3.38	1.73	2.13

表 4 在 SceneFlow 测试集不同区域的比较。

Method	Edges		Non-Edges	
	EPE	>1px	EPE	>1px
IGEV ^[38]	2.23	20.42	0.41	4.58
Selective-IGEV ^[35]	2.18	20.01	0.38	4.35
MonSter(我们的)	1.91	18.59	0.31	3.57

4.3 病态区域性能

MonSter 充分利用了单目深度先验的优势，有效克服了病态区域的挑战。为了验证这一点，我们在几个具有代表性的病态区域进行了比较，如反射区域、边缘和非边缘区域以及远距离背景：

- 反射区域 (Reflective Regions):** 我们在 KITTI 2012 基准的反射区域进行了比较。MonSter 在 KITTI 2012 排行榜上所有反射区域指标均排名第一。我们将现有的 SOTA 提升到了一个新的水平。在 Out-4(All) 指标上，MonSter 分别超过 Selective-IGEV 和 LoS 30.16% 和 29.24%。值得注意的是，与 SOTA 方法 NMRF-Stereo 相比，我们在 Out-3(All) 和 Out-4(All) 指标上分别实现了 58.32% 和 65.02% 的显著改进。
- 边缘与非边缘区域 (Edge & Non-edge Regions):** 立体匹配在边缘和低纹理区域面临挑战，我们的方法通过利用单目深度的优势解决了这个问题。为了在这些区域评估 MonSter，我们使用 Canny^[35]算子将 Scene Flow 测试集划分为边缘和非边缘区域。MonSter 在边缘区域和非边缘区域分别比基线^[38]优越 14.35% 和 24.39%。即使与专门设计用于解决边缘和无纹理区域的^[35]相比，我们的方法在两类区域的 EPE 指标上也分别实现了 12.39% 和 18.42% 的改进。
- 远距离背景 (Distant Backgrounds):** 立体匹配在远距离物体的深度感知方面存在困难，

我们通过结合 SGA 和 MGR 模块对其进行了改进。在 KITTI 2015 基准测试中，MonSter 在 D1-bg 指标上比基线提高了 18.12%。D1-bg 指标反映了专门针对背景区域平均的立体视差异常值的百分比。

表 5 零样本泛化基准。上半部分是仅在 SceneFlow 的泛化比较。下半部分比较了结合 Scene Flow、CREStereo 和 TartanAir 数据集训练模型的推广结果。**最好的结果加粗，次好的结果下划线标记。**

Method	KITTI-12 (>3px)	KITTI-15 (>3px)	Middlebury (2px)	ETH3D (1px)
Training Set	Scene Flow			
CFNet ^[29]	4.72	5.81	9.80	5.80
RAFT-Stereo ^[20]	5.12	5.74	9.36	<u>3.28</u>
CREStereo ^[16]	5.03	5.79	12.88	8.98
Selective-IGEV ^[35]	5.64	6.05	12.04	5.40
NMRF-Stereo ^[13]	<u>4.23</u>	<u>5.10</u>	7.54	3.82
IGEV ^[38]	4.84	5.51	<u>6.23</u>	3.62
MonSter(我们的)	3.62	3.97	5.17	2.03
Training Set	Scene Flow + CREStereo + Tartan Air			
IGEV ^[38]	3.95	4.30	5.56	2.38
MonSter(我们的)	2.95	3.23	2.94	1.21

4.4 零样本泛化

我们的方法不仅提高了域内数据集的准确性，还增强了泛化能力。我们评估了 MonSter 从合成数据集到未见过的真实世界场景的泛化性能。所有模型均在相同的合成数据集上训练，然后直接在真实世界数据集（KITTI^{[11][23]}、Middlebury^[27]和 ETH3D^[28]训练集）上进行测试。我们通过使用不同数据集进行训练进行了更彻底的泛化比较：

Scene Flow: 我们与仅在 Scene Flow 训练集上训练的 SOTA 方法进行了比较。MonSter 在四个数据集中实现了最佳的泛化性能。MonSter 甚至超过了 CFNet 和 CREStereo，这两种方法都是专门为跨域泛化设计的。

混合 3 个数据集：有趣的是，仅在训练集中添加 CREStereo 和 TartanAir, MonSter 的泛化能力就显著提高。这是因为 SceneFlow 数据集与真实世界分布差异很大，而 TartanAir 中的虚拟数据更类似于真实世界场景。这种相似性有助于我们的 SGA 和 MGR 模块更好地学习如何有效地整合单目和立体线索。如表 5 下半部分所示，我们在 Middlebury 和 ETH3D 上分别超过基线[38]47.12% 和 49.16%。

MonSter 强大的模拟到真实（sim-to-real）泛化能力令人鼓舞。未来，我们计划扩大模拟训练数据的规模和多样性，以进一步提升 MonSter 的性能。

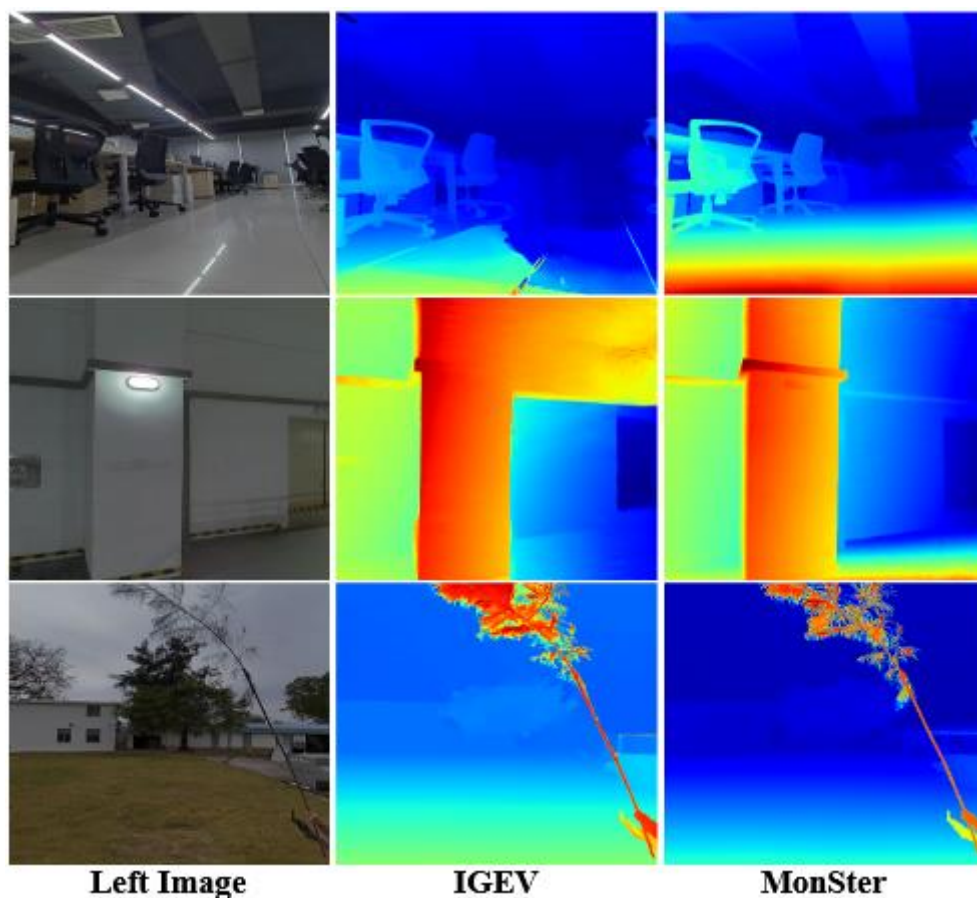


图 6 对我们采样的真实零样本数据进行可视化比较，提供了更全面的泛化能力评估。所有的模型都只在合成数据集 SceneFlow 上训练。Monster 在无纹理区域、反射区域和精细结构等方面的表现明显优于 IGEV

4.5 消融研究

为了证明 MonSter 各个组件的有效性，我们在 Scene Flow 上进行了消融实验，讨论了以下策略：

表 6 对拟议模块在 SceneFlow 测试集上的有效性进行消融研究。

Model	Monoculr Depth	Dispariy Fusion	Scale &Shift Refinement	Feature Sharig	EPE (px)	>1px (%)	Run-tim e(S)
Baseline (IGEV)					0.47	5.21	0.37
Mono+Conv	✓	Conv			0.46	5.12	0.64
Mono+MGR	✓	MGR	Conv		0.43	4.96	0.65
Mono+MGR+Conv	✓	MGR	SGA		0.42	4.82	0.65
Mono+MGR+SGA	✓	MGR	SGA	✓	0.39	4.43	0.66
Full mode(MonSter)	✓	MGR			0.37	4.25	0.64

 表 7 **Monster** 的有效性和通用性

Model	Mono Depth Model	Iteration Number	Sceneflow (EPE)	Run-time (S)
IGEV ^[38]		32	0.47	0.37
Full model	DepthAnythingV2[46]	32	0.37	0.64
Full model-4iter	DepthAnythingV2[46]	4	0.42	0.34
Full model-V1	DepthAnythingV1[45]	32	0.39	0.64
Full model-midas	MiDaS[25]	32	0.41	0.51

- 视差融合：** 单目深度通常包含大量噪声，这使得其与立体分支的视差融合变得复杂。为了证明我们的单目引导细化（MGR）融合方法的效率，我们将 MGR 模块替换为具有相同参数数量的基于卷积的沙漏网络，直接连接单目和立体视差进行融合，表示为 Mono+Conv。如表 6 所示，与“Mono+Conv”相比，“Mono+MGR”将 EPE 指标提高了 6.52%。
- 尺度和偏移细化：** 即使经过全局尺度-偏移对齐，单目深度仍然存在明显的尺度和偏移模糊。因此，利用立体分支的高置信度区域来校正单目视差的尺度和偏移至关重要。我们在“Mono+MGR”的基础上加入了立体引导细化模块（SGA）来优化尺度和偏移。如表 6 所示，与“Mono+MGR”相比，我们的 SGA 模块在 EPE 和 1 像素误差方面分别实现了 9.30% 和 10.69% 的改进，表明具有准确尺度和偏移的单目深度可以为立体匹配提供进一步的增强。同样，我们验证了我们的 SGA 模块比直接基于卷积的融合更有效。

与“Mono+MGR+Conv”相比，我们的 SGA 在 1 像素误差上仍实现了 8.09% 的改进。

- **特征共享：** 在大型数据集上预训练的单目深度估计模型的特征提取组件包含丰富的上下文信息，可以与立体分支共享。我们采用特征传输网络来映射这些特征，将其转换为与立体匹配兼容的表示。如表 6 最后两行所示，特征共享将 EPE 指标提高了 5.13%。
- **与其他单目模型的兼容性：** 为了验证 MonSter 对不同深度模型的通用性，我们将单目分支替换为^[45]的 ViT-large 版本和 MiDaS^[25]的 dpt_beit large 版本。所有 MonSter 变体均一致优于基线^[38]。这证明了 MonSter 的多功能性及其从未来单目深度模型进步中获益的潜力。
- **效率：** 我们使用 DepthAnything V2^[46]的 ViT-large 版本作为单目分支，这引入了时间和内存开销。与基线^[38]相比，我们的推理时间从 0.37s 增加到 0.64s。单目分支包含 335.3M 参数，立体分支有 12.6M 参数，SGA 和 MGR 模块有 8.2M 参数，我们的完整模型总共有 356.1M 参数。现有方法^{[1][36]}也使用基于 ViT 的编码器，参数数量很大。例如，CroCo-Stereo^[36]拥有 437.4M 参数，比 MonSter 大，但其性能较差，这表明参数数量并不决定最终效果。我们的研究侧重于准确性和泛化能力，鉴于准确性和泛化方面的显著改进（如表 2 和表 5 所示），额外的推理成本是可以接受的。值得注意的是，我们的方法简化了立体匹配，使我们仅需 4 次迭代 ($N_1 = N_2 = 2$) 即可实现 SOTA 性能。如表 7 所示，虽然基线需要 32 次迭代，但我们的方法仅需 4 次迭代即可实现 10.64% 的 EPE 改进，实现了更好的精度-速度权衡。可以通过编码器量化或蒸馏进一步减少内存，我们将其留作未来的工作。

5 结论

我们提出了 MonSter，这是一种新颖的流程，将立体匹配任务解耦为从相对深度恢复尺度和偏移的更简单范式。这种方法充分利用了单目方法提供的上下文和几何先验，同时避免了噪声和尺度模糊等问题。结果，我们的方法显著增强了立体匹配在病态区域的准确性和鲁棒性。MonSter 在五个广泛使用的基准上取得了比最先进技术实质性的改进。此外，MonSter 还表现出最佳的泛化性能，实验结果表明仍有相当大的改进空间。因此，我们未来的工作将侧重于扩大 MonSter 的规模，使其成为立体基础模型，从而支持需要度量深度的各种下游应用。

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MonSter: Marry **Monodepth** to **Stereo** Unleashes Power

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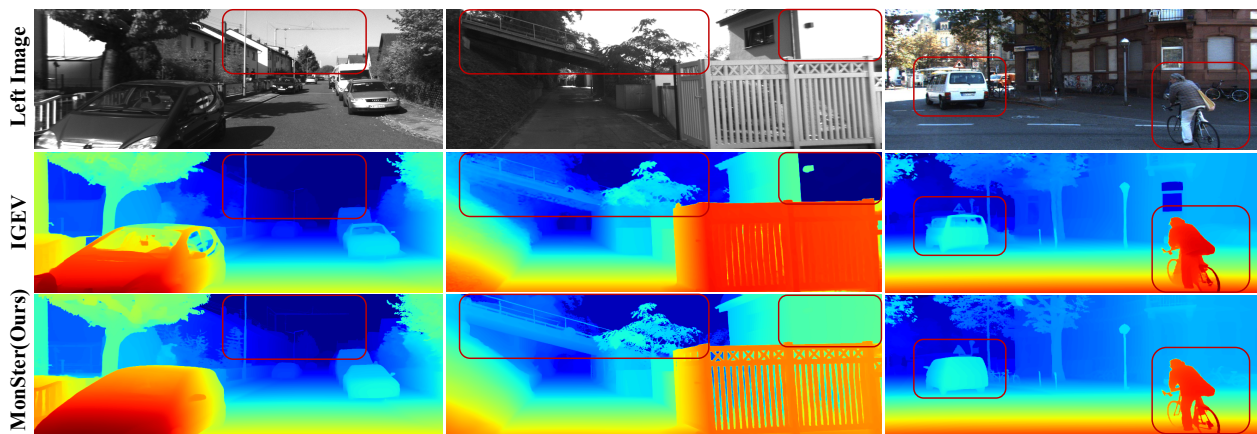


Figure 1. **Zero-shot generalization comparison:** all models are trained on Scene Flow and tested directly on KITTI. Compared to the baseline IGEV [37], our method MonSter shows significant improvement in challenging regions such as reflective surfaces, textureless areas, fine structures, and distant objects.

Abstract

Stereo matching recovers depth from image correspondences. Existing methods struggle to handle ill-posed regions with limited matching cues, such as occlusions and textureless areas. To address this, we propose MonSter, a novel method that leverages the complementary strengths of monocular depth estimation and stereo matching. MonSter integrates monocular depth and stereo matching into a dual-branch architecture to iteratively improve each other. Confidence-based guidance adaptively selects reliable stereo cues for monodepth scale-shift recovery. The refined monodepth is in turn guides stereo effectively at ill-posed regions. Such iterative mutual enhancement enables MonSter to evolve monodepth priors from coarse object-level structures to pixel-level geometry, fully unlocking the potential of stereo matching. As shown in Fig. 2, MonSter ranks 1st across five most commonly used leaderboards — SceneFlow, KITTI 2012, KITTI 2015, Middlebury, and

ETH3D. Achieving up to 49.5% improvements (Bad 1.0 on ETH3D) over the previous best method. Comprehensive analysis verifies the effectiveness of MonSter in ill-posed regions. In terms of zero-shot generalization, MonSter significantly and consistently outperforms state-of-the-art across the board. The code is publicly available at: <https://github.com/Junda24/MonSter>.

1. Introduction

Stereo matching estimates a disparity map from rectified stereo images, which can be subsequently converted into metric depth. It is the core of many applications such as self-driving, robotic navigation, and 3D reconstruction.

Deep learning based methods [5, 13, 15, 18, 33, 34, 39, 47, 49] have demonstrated promising performance on standard benchmarks. These methods can be roughly categorized into *cost filtering-based* methods and *iterative optimization-based* methods. Cost filtering-based methods [5, 13, 14, 20, 36, 40, 42] construct 3D/4D cost volume using CNN features, followed by a series of 2D/3D con-

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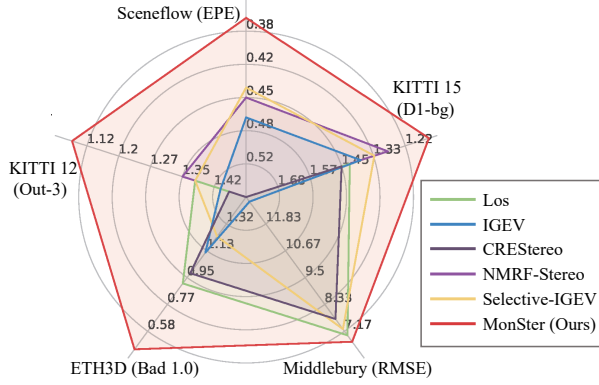


Figure 2. **Leaderboard performance.** Our method ranks 1st across 5 leaderboards, advancing SOTA by a large margin.

volutions for regularization and filtering to minimize mismatches. Iterative optimization-based methods [15, 18, 37, 48] initially construct an all-pairs correlation volume, then index a local cost to extract motion features, which guide the recurrent units (ConvGRUs) [9] to iteratively refine the disparity map.

Both types of methods essentially derive disparity from similarity matching, assuming visible correspondences in both images. This poses challenges in *ill-posed* regions with limited matching cues, e.g., occlusions, textureless areas, repetitive/thin structures, and distant objects with low pixel representation. Existing methods [5, 29, 31, 36, 41, 43, 48] address this issue by enhancing matching with stronger feature representations during feature extraction or cost aggregation: [17, 41] employ transformer as feature extractors; [36, 37] incorporate geometric information into the cost volume using attention mechanisms to strengthen matching information; [33, 48] further improve performance by introducing high-frequency information during the iterative refinement stage. However, these methods still struggle to fundamentally resolve the issue of mismatching, limiting their practical performance.

Unlike stereo matching, monocular depth estimation directly recovers 3D from a single image, which does not encounter the challenge of mismatching. While monocular depth provides complementary structural information for stereo, pre-trained models often yield relative depth with scale and shift ambiguities. As shown in Fig. 3, the prediction of monodepth models differ heavily from the ground-truth. Even after global scale and shift alignment, substantial errors still persist, complicating pixel-wise fusion of monocular depth and stereo disparity. Based on these insights, we propose *MonSter*, a novel approach that decomposes stereo matching into monocular depth estimation and per-pixel scale-shift recovery, which fully combines the strengths of monocular and stereo algorithms and overcomes the limitations from the lack of matching cues.

MonSter constructs separate branches for monocular

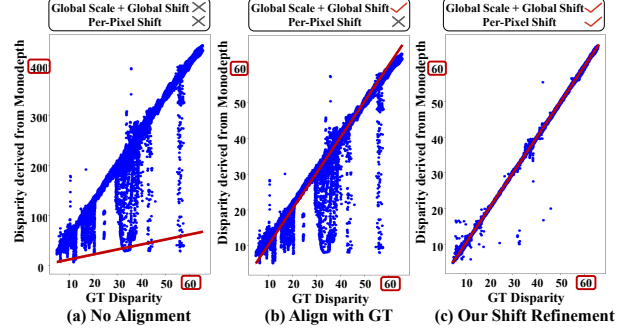


Figure 3. **Distance between GT disparity and the disparity derived from monocular depth [45] on KITTI dataset.** The red line indicates identical disparity maps. (a): Without any alignment. (b) Align depth with GT using global scale and global shift values (same for all pixels). (c) The monocular disparity produced by MonSter, with per-pixel shift refinement. Even globally aligned with GT, SOTA monocular depth models still exhibit severe noise. Our method MonSter effectively addresses this issue by refining monocular depth with per-pixel shift, which fully unlocks the power of monocular depth priors for stereo matching.

depth estimation and stereo matching, and adaptively fuses them through *Stereo Guided Alignment* (SGA) and *Mono Guided Refinement* (MGR) modules. SGA first rescales monodepth into a “monocular disparity” by aligning globally with stereo disparity. Then it uses condition-guided GRUs to adaptively select reliable stereo cues for updating the per-pixel monocular disparity shift. Symmetric to SGA, MGR uses the optimized monocular disparity as the condition to adaptively refine the stereo disparity in regions where matching fails. Through multiple iterations, the two branches effectively complement each other: 1) Though beneficial at coarse object-level, directly and unidirectionally fuse monodepth into stereo suffers from scale-shift ambiguities, which often introduces noise in complex regions such as slanted or curved surfaces. Refining monodepth with stereo resolves this issue effectively, ensuring the robustness of MonSter. 2) The refined monodepth in turn provides strong guidance to stereo in challenging regions. E.g., the depth perception ability of stereo matching degrades with distance due to smaller pixel proportions and the increased matching difficulty. Monodepth models pre-trained on large-scale datasets are less affected by such issues, which can effectively improve stereo disparity in the corresponding region.

Our main contributions can be summarized as follows:

- We propose a novel stereo-matching method MonSter, fully leveraging the pixel-level monocular depth priors to significantly improve the depth perception performance of stereo matching in ill-posed regions and fine structures.
- As shown in Fig. 2, MonSter ranks 1st across 5 widely-used leaderboards: KITTI 2012 [10], KITTI 2015 [21], Scene Flow [20], Middlebury [25] and ETH3D [26]. Advancing SOTA by up to 49.5%.

- Compared to SOTA methods, MonSter achieves the best zero-shot generalization consistently across diverse datasets. MonSter trained solely on synthetic data demonstrates strong performance across diverse real-world datasets (see Fig.1).

2. Related Work

Matching based Methods. Mainstream stereo matching methods recover disparity from matching costs. These methods can generally be divided into two categories: cost filtering-based methods and iterative optimization-based methods. Cost filtering-based methods [6, 7, 11, 13, 20, 27, 42] typically construct a 3D/4D cost volume using feature maps and subsequently employing 2D/3D CNN to filter the volume and derive the final disparity map. Constructing a cost volume with strong representational capacity and accurately regressing disparity from a noisy cost volume are key challenges. [5] proposes a stacked hourglass 3D CNN for better cost regularization. [13, 28, 36] propose the group-wise correlation volume, the attention concatenation volume, and the pyramid warping volume respectively to improve the expressiveness of the matching cost. Recently, a novel class of methods based on iterative optimization [3, 15, 18, 33, 37, 48] has achieved SOTA performance in both accuracy and efficiency. These methods use ConvGRUs to iteratively update the disparity by leveraging local cost values retrieved from the all-pairs correlation volume. Similarly, iterative optimization-based methods also primarily focus on improving the matching cost construction and the iterative optimization stages. CREStereo [15] proposes a cascaded recurrent network to update the disparity field in a coarse-to-fine manner. IGEV [37] proposes a Geometry Encoding Volume that encodes geometry and context information for a more robust matching cost. Both types of methods essentially derive depth from matching costs, which are inherently limited by ill-posed regions.

Stereo Matching with Structural Priors. Matching in ill-posed regions is challenging, previous methods tried to leverage structural priors to address this issue. EdgeStereo [29] enhances performance in edge regions by incorporating edge detection cues into the disparity estimation pipeline. SegStereo [43] utilizes semantic cues from a segmentation network as guidance for stereo matching, improving performance in textureless regions. However, semantic and edge cues only provide object-level priors, which are insufficient for pixel-level depth perception. Consequently, they are not effective in challenging scenes with large curved or slanted surfaces. Therefore, some methods incorporate monocular depth priors to provide per-pixel guidance for stereo matching through dense relative depth. CLStereo [46] introduces a monocular branch that serves as a contextual constraint, transferring geometric priors from the monocular to the stereo branch. Los [16] uses monocular depth as local

structural prior to generating the slant plane, which can explicitly leverage structure information for updating disparities. However, monocular depth estimation suffers from severe scale and shift ambiguities as shown in Fig.3. Directly using it as a structural prior to constrain stereo matching can introduce heavy noise. Our MonSter adaptively selects reliable stereo disparity to correct the scale and shift of monodepth, which fully leverages the monocular depth priors while avoiding noise, thereby significantly enhancing stereo performance in ill-posed regions.

3. Method

As shown in Fig. 4, MonSter consists of 1) a *monocular depth branch*, 2) a *stereo matching branch* and 3) a *mutual refinement module*. The two branches estimate initial monocular depth and the stereo disparity, which are fed into mutual refinement to iteratively improve each other.

3.1. Monocular and Stereo Branches

The monocular depth branch can leverage most monocular depth models to achieve non-trivial performance improvements (see Sec.4.5 for analysis). Our best empirical configuration uses pretrained DepthAnythingV2 [45] as the monocular depth branch, which uses DINOv2 [22] as the encoder and DPT [24] as the decoder. The stereo matching branch follows IGEV [37] to obtain the initial stereo disp, with modifications only to the feature extraction component, as shown in Fig. 4. To efficiently and fully leverage the pretrained monocular model, the stereo branch shares the ViT encoder in DINOv2 with the monocular branch, with parameters frozen to prevent the stereo matching training from affecting its generalization ability. Moreover, the ViT architecture extracts feature at a single resolution, while recent stereo matching methods commonly utilize multi-scale features at four scales (1/32, 1/16, 1/8, and 1/4 of the original image resolution). To fully align with IGEV, we employ a stack of 2D convolutional layers, denoted as the feature transfer network, to downsample and transform the ViT feature into a collection of pyramid features $\mathcal{F} = \{F_0, F_1, F_2, F_3\}$, where $F_k \in \mathbb{R}^{\frac{H}{2^{5-k}} \times \frac{W}{2^{5-k}} \times c_k}$. We follow IGEV to construct Geometry Encoding Volume and use the same ConvGRUs for iterative optimization. To balance accuracy and efficiency, we perform only N_1 iterations to obtain a initial stereo disparity with reasonable quality.

3.2. Mutual Refinement

Once the initial (relative) monocular depth and stereo disparity are obtained, they are fed into the mutual refinement module to iteratively refine each other. We first perform *global scale-shift alignment*, which converts monocular depth into a disparity map and aligns it coarsely with stereo outputs. Then we iteratively perform a dual-branched refinement: *Stereo guided alignment (SGA)* leverages stereo

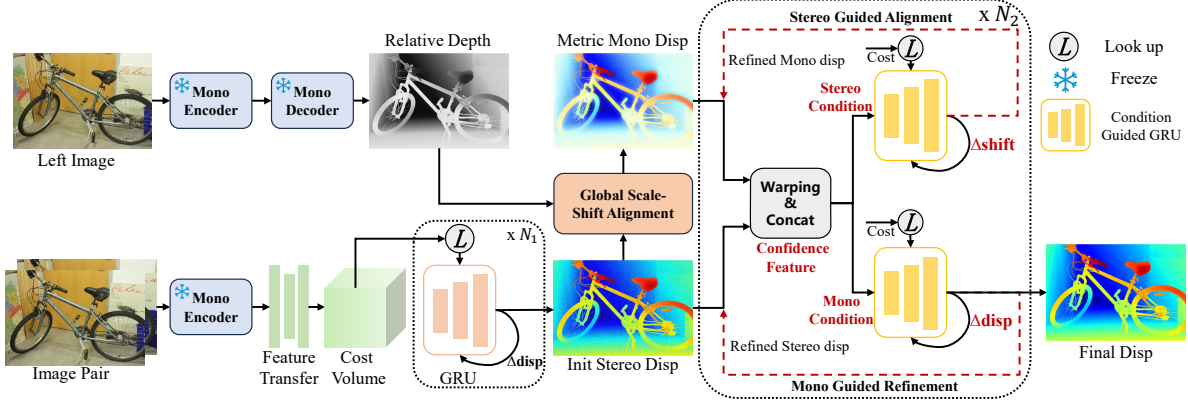


Figure 4. **Overview of MonSter.** MonSter consists of a monocular depth estimation branch, a stereo matching branch, and a mutual refinement module. It iteratively improves one branch with priors from the other, effectively resolving the ill-posedness in stereo matching.

cues to update the per-pixel shift of the monocular disparity; *Mono guided refinement (MGR)* leverages the aligned monocular prior to further refine stereo disparity.

Global Scale-Shift Alignment. Global Scale-Shift Alignment performs least squares optimization over a global scale s_G and a global shift t_G to coarsely align the inverse monocular depth with the stereo disparity:

$$s_G, t_G = \arg \min_{s_G, t_G} \sum_{i \in \Omega} (s_G D_M(i) + t_G - D_S^0(i))^2 \quad (1)$$

$$D_M^0 = s_G D_M + t_G,$$

where $D_M(i)$ and $D_S^0(i)$ are the inverse monocular depth and stereo disparity at the i -th pixel. Ω represents the region where stereo disparity values fall between 20% to 90% sorting in ascending order, which helps to filter unreliable regions such as the sky, extremely distant areas, and close-range outliers. Intuitively, this step converts the inverse monocular depth into a disparity map coarsely aligned with the stereo disparity, enabling effective mutual refinement in the same space. We call this aligned disparity map D_M^0 the *monocular disparity* in the remainder of the paper.

Stereo Guided Alignment (SGA). Though coarsely aligned, a unified shift t_G is not sufficient to recover accurate monocular disparity at different pixels. To fully release the potential of monocular depth prior, SGA leverages intermediate stereo cues to further estimate a per-pixel residual shift. To avoid introducing noisy stereo cues, SGA uses *confidence based guidance*. In each update step j , we compute the confidence using the flow residual map F_S^j , which is obtained by warping and subtracting features based on the stereo disparity D_S^j as:

$$F_S^j(x, y) = \left\| F_3^L(x, y) - F_3^R(x - D_S^j, y) \right\|_1 \quad (2)$$

where F_3^L, F_3^R represent the quarter-resolution features of left and right images in $\mathcal{F}$ respectively. For each iteration, we also use the current stereo disparity D_S^j to index from

the Geometry Encoding Volume to obtain geometry features of stereo branch G_S^j follow IGEV. Concatenated with F_S^j and D_S^j , we obtain the stereo condition as:

$$x_S^j = [\text{En}_g([G_S^j, F_S^j, D_S^j]), \text{En}_d(D_M^j, D_M^j)], \quad (3)$$

where En_g and En_d are two convolutional layers for feature encoding. We feed x_S^j into condition-guided ConvGRUs to update the hidden state h_m^{j-1} of monocular branch as:

$$\begin{aligned} z^j &= \sigma(\text{Conv}([h_M^{j-1}, x_S^j], W_z) + c_k), \\ r^j &= \sigma(\text{Conv}([h_M^{j-1}, x_S^j], W_r) + c_r), \\ \tilde{h}_M^j &= \tanh(\text{Conv}([r^j \odot h_M^{j-1}, x_S^j], W_h) + c_h), \\ h_M^j &= (1 - z^j) \odot h_M^{j-1} + z^j \odot \tilde{h}_M^j, \end{aligned} \quad (4)$$

where c_k, c_r, c_h are context features. Based on the hidden state h_M^j , we decode a *residual shift* Δt through two convolutional layers to update the monocular disparity:

$$D_M^{j+1} = D_M^j + \Delta t. \quad (5)$$

Mono Guided Refinement (MGR). Symmetric to SGA, MGR leverages the aligned monocular disparity to address stereo deficiencies in ill-posed regions, thin structures, and distant objects. Specifically, we employ the same condition-guided GRU architecture with independent parameters to refine stereo disparity. We simultaneously calculate the flow residual maps F_M^j, F_S^j and geometric features G_M^j, G_S^j for both the monocular and stereo branches, providing a comprehensive stereo refinement guidance:

$$\begin{aligned} F_M^j(x, y) &= \left\| F_3^L(x, y) - F_3^R(x - D_M^j, y) \right\|_1, \\ x_M^j &= [\text{En}_g([G_M^j, F_M^j, D_M^j]), \text{En}_d(D_M^j, D_M^j)], \\ x_S^j &= [\text{En}_g([G_S^j, F_S^j, D_S^j]), \text{En}_d(D_S^j, D_S^j)], \end{aligned} \quad (6)$$

where G_M^j is the geometry features of the monocular branch obtained by indexing from the Geometry Encoding

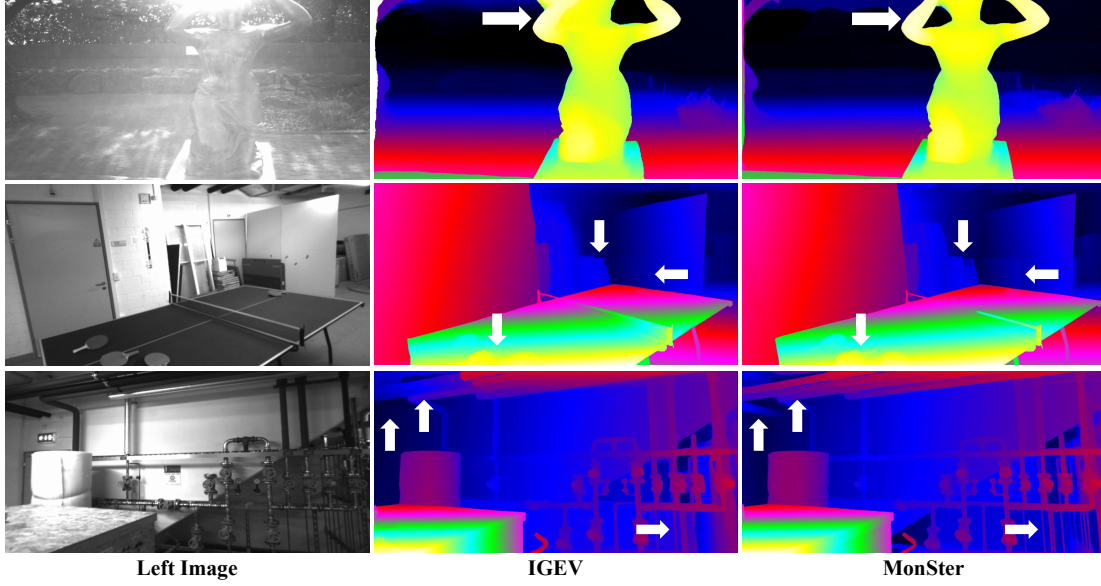


Figure 5. **Qualitative results on ETH3D.** MonSter outperforms IGEV in challenging areas with strong reflectance, fine structures etc.

Volume using the monocular disparity D_M^j . x_M^j represents the monocular condition for ConvGRUs, and we use the Eq. (4) to update the hidden state h_S^i similarly, with only the condition input changed from x_S^i to x_M^i . We use the same two convolutional layers to decode the residual disparity Δd and update the current stereo disparity following Eq. (5). After N_2 rounds of dual-branched refinement, the disparity of the stereo branch is the final output of MonSter.

3.3. Loss Function

We use the L1 loss to supervise the output from two branches. We denote the set of disparities from the first N_1 iterations of the stereo branch as $\{d_i\}_{i=0}^{N_1-1}$ and follow [18] to exponentially increase the weights as the number of iterations increases. The total loss is defined as the sum of the monocular branch loss $\mathcal{L}_{Mono}$ and the stereo branch loss $\mathcal{L}_{Stereo}$ as follows:

$$\begin{aligned} \mathcal{L}_{Stereo} &= \sum_{i=0}^{N_1-1} \gamma^{N_1+N_2-i} \|d_i - d_{gt}\|_1 + \\ &\quad \sum_{i=N_1}^{N_1+N_2-1} \gamma^{N_1+N_2-i} \|D_S^{i-N_1} - d_{gt}\|_1, \quad (7) \\ \mathcal{L}_{Mono} &= \sum_{i=N_1}^{N_1+N_2-1} \gamma^{N_1+N_2-i} \|D_m^{i-N_1} - d_{gt}\|_1 \end{aligned}$$

where $\gamma = 0.9$, and d_{gt} is the ground truth.

4. Experiment

4.1. Implementation Details

We implement MonSter with PyTorch and perform experiments using NVIDIA RTX 3090 GPUs. We use the

AdamW [19] optimizer and clip gradients to the range $[-1, 1]$ following baseline [37]. We use the one-cycle learning rate schedule with a learning rate of $2e-4$ and train MonSter with a batch size of 8 for 200k steps as the pretrained model. For the monocular branch, we use the ViT-large version of DepthAnythingV2 [45] and freeze its parameters to prevent training of stereo-matching tasks from affecting its generalization and accuracy.

Following the standard [15, 16, 33], we pretrain MonSter on Scene Flow [20] for most experiments. For fine-tuning on ETH3D and Middlebury, we follow SOTA methods [15, 16, 33] to create the Basic Training Set (BTS) from various public datasets for pretraining, including Scene Flow [20], CREStereo [15], Tartan Air [32], Sintel Stereo [4], FallingThings [30] and InStereo2k [2].

4.2. Benchmark Performance

To demonstrate the outstanding performance of our method, we evaluate MonSter on the five most commonly used benchmarks: KITTI 2012 [10], KITTI 2015 [21], ETH3D [26], Middlebury [25] and Scene Flow [20].

Scene Flow [20]. As shown in Tab. 1, we achieve a new state-of-the-art performance with an EPE metric of 0.37 on Scene Flow, surpassing our baseline [37] by 21.28% and outperforming the SOTA method [33] by 15.91%.

ETH3D [26]. Following the SOTA methods [15, 16, 33, 38, 41], the full training set is composed of the BTS and ETH3D training set. Our MonSter ranks 1st on the ETH3D leaderboard. As shown in Tab. 2, MonSter achieves the best performance among all published methods for all metrics. Compared with baseline IGEV, we achieve improvements of 58.93%, 52.32%, and 41.18% in the three reported metrics, respectively. Qualitative comparisons shown in Fig.5

Method	GwcNet[13]	LEAStereo[8]	ACVNet[36]	IGEV[37]	Selective-IGEV[33]	MonSter (Ours)
EPE (px)↓	0.76	0.78	0.48	0.47	<u>0.44</u>	0.37 (−15.91%)

Table 1. **Quantitative evaluation on Scene Flow test set.** The **best** result is bolded, and the second-best result is underscored.

Method	ETH3D[26]			Middlebury[25]			KITTI 2015[10]				KITTI 2012[10]			
	Bad1.0	Bad1.0	RMSE	Bad2.0	Bad2.0	RMSE	D1-bg	D1-all	D1-bg	D1-all	Out-2	Out-2	Out-3	Out-3
	Noc	All	Noc	Noc	All	Noc	Noc	Noc	All	All	Noc	All	Noc	All
GwcNet[13]	6.42	6.95	0.69	-	-	-	1.61	1.92	1.74	2.11	2.16	2.71	1.32	1.70
GANet[47]	6.22	6.86	0.75	-	-	-	1.40	1.73	1.55	1.93	1.89	2.50	1.19	1.60
LEAStereo[8]	-	-	-	7.15	12.10	8.11	1.29	1.51	1.40	1.65	1.90	2.39	1.13	1.45
ACVNet[36]	2.58	2.86	0.45	13.70	19.50	32.2	1.26	1.52	1.37	1.65	1.83	2.35	1.13	1.47
RAFT-Stereo[18]	2.44	2.60	0.36	4.74	9.37	8.41	1.44	1.69	1.58	1.82	1.92	2.42	1.30	1.66
CREStereo[15]	0.98	1.09	0.28	3.71	8.13	7.70	1.33	1.54	1.45	1.69	1.72	2.18	1.14	1.46
IGEV[37]	1.12	1.51	0.34	4.83	8.16	12.80	1.27	1.49	1.38	1.59	1.71	2.17	1.12	1.44
CroCo-Stereo[35]	0.99	1.14	0.30	7.29	11.11	8.91	1.30	1.51	1.38	1.59	-	-	-	-
DLNR[48]	-	-	-	3.20	6.98	7.78	1.42	1.61	1.60	1.76	-	-	-	-
Selective-IGEV[33]	1.23	1.56	0.29	2.51	6.04	7.26	1.22	1.44	1.33	1.55	1.59	2.05	1.07	1.38
LoS[16]	0.91	1.03	0.31	4.20	8.03	6.99	1.29	1.52	1.42	1.65	1.69	2.12	1.10	1.38
NMRF-Stereo[12]	-	-	-	-	-	-	1.18	1.46	1.28	1.57	1.59	2.07	1.01	1.35
MonSter(Ours)	0.46	0.72	0.20	2.64	6.14	6.71	1.05	1.33	1.13	1.41	1.36	1.75	0.84	1.09

Table 2. **Results on four popular benchmarks.** All results are derived from official leaderboard publications or corresponding papers. All metrics are presented in percentages, except for RMSE, which is reported in pixels. For testing masks, “All” denotes being tested with all pixels while “Noc” denotes being tested with a non-occlusion mask. The **best** and second best are marked with colors.

exhibit a similar trend. Notably, even compared with the previous best method LoS, we improved the Bad 1.0 (Noc) metric from 0.91 to 0.46, achieving a 49.45% improvement.

Middlebury. Also following [15, 16, 33, 41], the training set of Middlebury is the combination of the BTS and Middlebury training set. As shown in Tab. 2, MonSter outperforms all existing methods in terms of RMSE metric. Compared with baseline [37], we achieved an improvement of 45.34% on the Bad 2.0 (Noc) metric.

KITTI. Following the training of SOTA methods [12, 15, 33], we finetuned Scene Flow pretrained model on the mixed dataset of KITTI 2012 and KITTI 2015 for 50k steps. At the time of writing, MonSter ranks 1st simultaneously on both the KITTI 2012 and KITTI 2015 leaderboards. As shown in Tab. 2, we achieve the best performance for all metrics. On KITTI 2015, MonSter surpasses CREStereo [15] and Selective-IGEV [33] by 16.57% and 9.03% on D1-all metric of all regions, respectively. As for KITTI 2012, we significantly outperform the existing SOTA by a large margin. Compared to the current highest-precision method, NMRF-Stereo [12] and Selective-IGEV [33], we achieve a 19.26% improvement and a 21.01% improvement respectively in the Out-3 metric across all regions.

4.3. Performance in Ill-posed Regions

MonSter fully leverages the advantages of monocular depth priors, effectively overcoming challenges in ill-posed regions. To validate this, we conduct comparisons on several representative ill-posed regions, such as reflective ar-

Method	KITTI 2012 Reflective Region					
	Out-2	Out-2	Out-3	Out-3	Out-4	Out-4
	Noc	All	Noc	All	Noc	All
ACVNet[36]	11.42	13.53	7.03	8.67	5.18	6.48
CREStereo[15]	9.71	11.26	6.27	7.27	4.93	5.55
IGEV[37]	7.57	8.80	4.35	5.00	3.16	3.57
Selective-IGEV[33]	6.73	7.84	3.79	4.38	2.66	3.05
LoS[16]	6.31	7.84	3.47	4.45	2.41	3.01
NMRF-Stereo[12]	10.02	12.34	6.35	8.11	4.80	6.09
MonSter(Ours)	5.66	6.81	2.75	3.38	1.73	2.13

Table 3. **Results of the reflective regions on KITTI 2012 leaderboard.** The **best** and second best are marked with colors.

Method	Edges		Non-Edges	
	EPE	>1px	EPE	>1px
IGEV [37]	2.23	20.42	0.41	4.58
Selective-IGEV [33]	2.18	20.01	0.38	4.35
MonSter(Ours)	1.91	18.59	0.31	3.57

Table 4. **Comparison on SceneFlow test set in different regions.**

eas, edge and non-edge regions, and distant backgrounds:

- **Reflective Regions:** We conducted comparisons on the reflective regions of KITTI 2012 benchmark. MonSter ranks 1st on the KITTI 2012 leaderboard for all metrics of reflective regions. As shown in Tab.3, we have elevated the existing SOTA to a new level. MonSter surpasses Selective-IGEV and LoS by 30.16% and 29.24% on Out-4(All) metric, respectively. Notably, compared to the SOTA method NMRF-Stereo, we achieved significant

Method	KITTI-12 (>3px)	KITTI-15 (>3px)	Middlebury (>2px)	ETH3D (>1px)
Training Set		Scene Flow		
CFNet[27]	4.72	5.81	9.80	5.80
RAFT-Stereo[18]	5.12	5.74	9.36	3.28
CREStereo[15]	5.03	5.79	12.88	8.98
Selective-IGEV[33]	5.64	6.05	12.04	5.40
NMRF-Stereo[12]	4.23	5.10	7.54	3.82
IGEV[37]	4.84	5.51	6.23	3.62
MonSter(Ours)	3.62	3.97	5.17	2.03
Training Set		Scene Flow + CREStereo + Tartan Air		
IGEV[37]	3.95	4.30	5.56	2.38
MonSter(Ours)	2.95	3.23	2.94	1.21

Table 5. **Zero-shot generalization benchmark.** The top half is generalization comparisons only trained on Scene Flow. The bottom half compares generalizations for models trained on a combination of Scene Flow, CREStereo, and TartanAir datasets.

improvements of 58.32% and 65.02% in the Out-3(All) and Out-4(All) metrics respectively.

- **Edge & Non-edge Regions:** Stereo matching faces challenges in edge and low-texture regions, which our method addresses by leveraging the strengths of monocular depth. To evaluate MonSter in these areas, we divide the Scene Flow test set into edge and non-edge regions using the Canny operator following [33]. As shown in Tab. 4, MonSter outperforms the baseline [37] by 14.35% and 24.39% in edge regions and non-edge regions. Even when compared to [33], which is specifically designed to address edge and textureless regions, our method achieves improvements of 12.39% and 18.42% in two types of regions on EPE metric, respectively.
- **Distant Backgrounds:** Stereo matching struggles with depth perception for distant objects, we improve it by incorporating our SGA and MGR modules. As shown in Tab.2, MonSter improves the D1-bg metric by 18.12% compared to the baseline on KITTI 2015 benchmark. The D1-bg metric reflects the percentage of stereo disparity outliers averaged specifically over background regions.

4.4. Zero-shot Generalization

Our method not only improves the accuracy of in-domain datasets but also enhances generalization ability. We evaluate the generalization performance of MonSter from synthetic datasets to unseen real-world scenes. All models were trained on the same synthetic datasets and then tested directly on real-world datasets: KITTI[10, 21], Middlebury[25], and ETH3D[26] training sets. We conducted a more thorough generalization comparison by training with different datasets:

Scene Flow: We compared with SOTA methods which are only trained on the Scene Flow training set. As shown in the top half of Tab. 5, MonSter achieves the best gener-

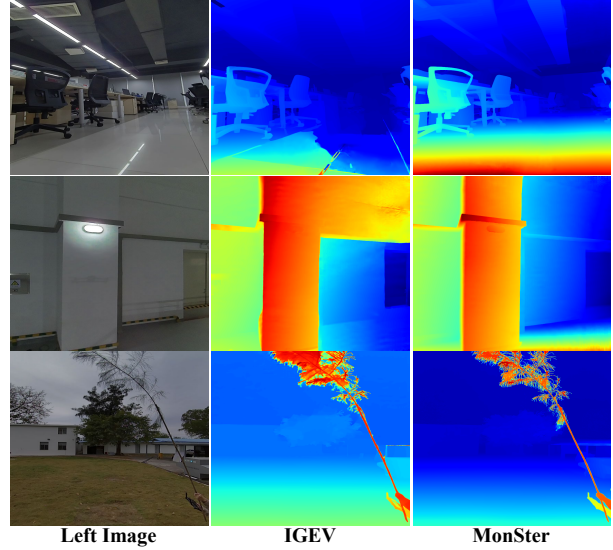


Figure 6. **Visual comparisons on our captured real-world zero-shot data provide a more comprehensive assessment of generalization capability.** All models are trained solely on the synthetic SceneFlow dataset. MonSter significantly outperforms IGEV in textureless regions, reflective areas and fine structures, etc.

alization performance across four datasets. MonSter even surpasses CFNet and CREStereo, both of which are specifically designed for cross-domain generalization.

Mix 3 datasets: Interestingly, only adding CREStereo and TartanAir to the training set, the generalization of MonSter improves significantly. This is because the SceneFlow dataset differs substantially from real-world distributions, whereas the virtual data in TartanAir more closely resemble real-world scenes. This similarity helps our SGA and MGR modules better in learning how to effectively integrate monocular and stereo cues. As shown in the bottom half of Tab. 5, we surpass the baseline[37] by 47.12% and 49.16% on Middlebury and ETH3D respectively.

The strong sim-to-real generalization of MonSter is highly encouraging. In the future, we plan to scale up the size and diversity of simulation training data to further boost the performance of MonSter.

4.5. Ablation Study

To demonstrate the effectiveness of each component of MonSter, we ablation on Scene Flow [20] and the following strategies are discussed:

- **Disparity Fusion:** Monocular depth often carries substantial noise, which complicates its fusion with the stereo branch’s disparity. To demonstrate the efficiency of our Mono Guided Refinement (MGR) fusion method, we replaced the MGR module with a convolution-based hourglass network of equal parameter numbers, directly concatenating the monocular and stereo disparities for fusion, denoted as ‘Mono+Conv’. As shown in Tab.6, compared with ‘Mono+Conv’, ‘Mono+MGR’ improves the

Model	Monocular Depth	Disparity Fusion	Scale & Shift Refinement	Feature Sharing	EPE (px)	>1px (%)	Run-time (S)
Baseline (IGEV)					0.47	5.21	0.37
Mono+Conv	✓	Conv			0.46	5.12	0.64
Mono+MGR	✓	MGR			0.43	4.96	0.65
Mono+MGR+Conv	✓	MGR	Conv		0.42	4.82	0.65
Mono+MGR+SGA	✓	MGR	SGA		0.39	4.43	0.66
Full model (MonSter)	✓	MGR	SGA	✓	0.37	4.25	0.64

Table 6. Ablation study of the effectiveness of proposed modules on the Scene Flow test set.

Model	Mono Depth Model	Iteration Number	SceneFlow (EPE)	Run-time (S)
IGEV[37]		32	0.47	0.37
Full model	DepthAnythingV2[45]	32	0.37	0.64
Full model-4iter	DepthAnythingV2[45]	4	0.42	0.34
Full model-V1	DepthAnythingV1[44]	32	0.39	0.64
Full model-midas	MiDaS[23]	32	0.41	0.51

Table 7. Efficiency and universality of MonSter.

EPE metric by 6.52%.

- **Scale & Shift Refinement:** As shown in Fig.3, even after global scale-shift alignment, the monocular depth still suffers from significant scale and shift ambiguities. Therefore, leveraging high-confidence regions from the stereo branch to correct the scale and shift of the monocular disparity is essential. We incorporated our Stereo Guided Refinement module (SGA) to optimize the scale and shift based on ‘Mono+MGR’. As shown in Tab.6, compared with ‘Mono+MGR’, our SGA module achieves improvements of 9.30% in EPE and 10.69% in 1-pixel error, demonstrating that monocular depth with accurate scale and shift can provide further enhancements to stereo matching. Similarly, we validated that our SGA module is more effective than direct convolution-based fusion. Compared to ‘Mono+MGR+Conv’, our SGA still achieves an 8.09% improvement in 1-pixel error.
- **Feature Sharing:** The feature extraction component of monocular depth estimation models, pre-trained on large datasets, contains rich contextual information that can be shared with the stereo branch. We employ a feature transfer network to map these features, transforming them into a representation compatible with stereo matching. As shown in the last 2 rows of Tab.6, feature sharing improved the EPE metric by 5.13%.

Compatibility with Other Monocular Models. To validate the generalizability of MonSter to different depth models, we replace the monocular branch with the ViT-large version of [44] and the dpt.beit.large version of MiDaS [23]. As shown in Tab.7, all MonSter variants consistently outperform the baseline [37]. This demonstrates the versatility of MonSter and its potential to benefit from future advancements in monocular depth models.

Efficiency. We use the ViT-large version of DepthAnythingV2 [45] as the monocular branch, which introduces

time and memory overheads. Compared to the baseline [37], our inference time increases from 0.37s to 0.64s. With the monocular branch containing 335.3M parameters, the stereo branch having 12.6M parameters, and the SGA and MGR modules having 8.2M parameters, our full model has a total of 356.1M parameters. Existing methods [1, 35] also use ViT-based encoders with a large number of parameters. For example, CroCo-Stereo [35] has 437.4M parameters, larger than MonSter, yet its performance is inferior, as shown in Tab.2, this demonstrates that the number of parameters does not determine the final effectiveness. Our research focuses on the accuracy and generalization capabilities, given the significant improvement in accuracy and generalization (as shown in Tab.2 and Tab.5), the additional inference cost is acceptable. Notably, our approach simplifies stereo matching and enables us to achieve SOTA performance with only 4 iterations ($N_1=N_2=2$). As shown in Tab.7, while the baseline requires 32 iterations, our method achieves a 10.64% improvement in EPE with just 4 iterations, achieving a better accuracy-speed trade-off. Further memory reduction can be achieved through encoder quantization or distillation, which we leave as future work.

5. Conclusion

We propose MonSter, a novel pipeline that decouples the stereo matching task into a simpler paradigm of recovering scale and shift from relative depth. This approach fully leverages the contextual and geometric priors provided by monocular methods while avoiding issues such as noise and scale ambiguity. As a result, our method significantly enhances the accuracy and robustness of stereo matching in ill-posed regions. MonSter achieves substantial improvements over the state-of-the-art on five widely used benchmarks. Additionally, MonSter also demonstrates the best generalization performance, and experimental results show that there is still considerable room for improvement. Therefore, our future work will focus on scaling up MonSter to serve as a stereo foundation model, enabling a wide range of downstream applications that require metric depth.

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